Jasper **Blank**

Control & Embedded Systems · Medical Devices & Robotics
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Profile

Biomedical engineer finishing an MSc in Biorobotics at the University of Twente, building a surgical end-effector for my thesis and, before it, taking a patient-warming device from open-loop to closed-loop control. I work across the boundary between simulation, firmware and test.

Experience

TSC Life Sciences — *Amersfoort, NL*

R&D Intern · Closed-loop control for the Mistral Air MA1200 surgical warming system · graded 8/10 — *Feb – May 2026*

- Replaced open-loop operation with a **two-layer cascade controller**: fast inner loop on skin temperature, slow outer loop correcting the setpoint from a core-temperature estimate.
- Derived an **LTI reduction of a 23-state Fiala thermoregulation model**, validated to 0.05 °C RMS against the non-linear reference, and used it as the design plant.
- Built a **model predictive controller** in simulation and benchmarked it against the tuned PID over a full surgery scenario.
- Wrote and flashed **custom firmware** over UPDI, tracing the ADC path through an unfamiliar legacy codebase to read two thermistors added on unused board channels.
- Validated to the **IEC 60601-2-35 Annex CC** thermal-load procedure: 182 minutes closed-loop, outlet held at $\sigma = 0.030$ °C.
- Designed an **isolated thermistor front-end** to IEC 60601-1 Type BF (2× MOPP, 4 kV, 8 mm creepage) with a costed bill of materials.

Z.P.V. Piranha — *Enschede, NL*

Treasurer — *Jul 2023 – Sep 2024*

- Budgeting, bookkeeping and financial reporting for a 400-member association with annual cashflow above €50,000 (Davilex, Visma).
- Board member for strategic and operational decisions, reporting to members and external stakeholders.

University of Twente — *Enschede, NL*

Swimming Instructor & Lifeguard · sole responsibility for pool safety on unsupervised shifts — *Mar 2023 – present*

Education

University of Twente — *Enschede, NL*

MSc Biorobotics — *2024 – present*

- **Thesis (in progress), Surgical Robotics Lab:** an Ø8 mm dual-endoscope probe for needle-based vessel targeting. Built the probe, PDMS phantoms, stereo vision and motor control, with a first guarded approach to a vessel on hardware. In a digital twin, two cameras reached the target in 20/20 closed-loop runs, one camera in 0/20; physical trials next.

University of Twente — *Enschede, NL*

BSc Biomedical Engineering — *2019 – 2024*

- Final project: designed and tested miniature low-cracking-pressure check valves for an incubator-compatible pump intended for blood-mimicking in-vitro flow models. SolidWorks, actuation and flow-rate control.

Projects

Epistack — *jasperblank.com/epistack*

Auditable claim-source mapping

- Decomposes a confidence figure into a reader-owned prior plus source-grounded updates, so every shift traces to a verbatim quotation and a stated reason. Modular Python pipeline with offline tests.

Game jams — *jasperblank.itch.io*

Eight published browser games

- Released across GameMaker and Godot, several built inside 72-hour jams — solo and with a recurring art collaborator, taking the technical development role. **Two third-place jam finishes.** AI assistance disclosed per project.

1D Chess — *jasperblank.com/1d-chess*

Search engine and solver

- Game-tree search with transposition tables, killer-move ordering and late move reduction, benchmarked against MCTS.

Skills

Control & Modelling	PID and cascade control, model predictive control, Kalman filtering, system identification, state-space modelling, constrained optimisation (SLSQP)
Perception	Stereo vision and depth resolution, camera calibration (AprilTag), signal processing, closed-loop visual servoing
Software	Python, C/C++, MATLAB & Simulink, JavaScript, GDScript · Git, Godot, LaTeX · AI coding agents
Hardware	Embedded firmware, microcontroller ADC and sensor interfacing, schematic reading, mechatronics prototyping, SolidWorks · IEC 60601-1 and -2-35 familiarity (not a certification)
Other	BHV (emergency response), Red Cross first aid, open-water lifeguard · Dutch (native), English (fluent)